

MATLAB SCRIPT

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%-----  
% Static Characteristics of your gimbal motor  
%-----  
  
wg = Mean_Rate_Gyro * 180 / pi ; % mean of rate gyro outputs [deg/s]  
Vm = Mean_Motor_Volt ;           % mean of motor driving voltage [V]  
  
%-----  
% Linearization Function Design  
%-----  
  
% Motor Angular Speed  
  
MAX_RATE = 1400; % [deg/s]  
MIN_RATE = 30; % [deg/s]  
  
% Extract Data Points for CW and CCW Directions  
  
[~,idxN1]= min(abs(wg - (-MAX_RATE))) ; % CCW direction  
[~,idxN2]= min(abs(wg - (-MIN_RATE))) ;  
  
[~,idxP1]= min(abs(wg - MIN_RATE))      ; % CW direction  
[~,idxP2]= min(abs(wg - MAX_RATE))      ;  
  
% You can design the linearization function by applying the basic fitting  
% function to the following figures  
% It is highly recommended to choose the 4th order polynomial  
% whose coefficients has 6 significant digits  
  
% figure, plot(wg(idxP1:idxP2), Vm(idxP1:idxP2)) ; grid on, box on  
% figure, plot(wg(idxN1:idxN2), Vm(idxN1:idxN2)) ; grid on, box on  
  
% Examples of linearization functions for each direction  
  
CoefP = [6.0884e-13;-9.8931e-10; 5.8845e-07; 0.00069526; 2.7784] ;  
CoefN = [-4.5324e-13;-5.4804e-10;-1.9035e-07;0.00087339; 2.2991] ;  
  
%-----  
% Validity Check of Linearization Algorithm  
%-----  
  
bufwc = -1500:1:1500 ; % control command of a stabilization loop  
N = length(bufwc) ;  
  
wc_DZ = 20.0 ; % desired dead-zone [deg/s]  
wc_SAT = 1400.0 ; % desired saturation [deg/s]  
  
for idx = 1:N  
    wc = bufwc(idx) ; % control command
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%-----
% LINEARIZATION FUNCTION
%-----

if( wc >= wc_DZ ) % CW

    wc = min( wc, wc_SAT ) ;
    H = [ wc^4 ; wc^3 ; wc^2 ; wc ; 1 ] ;

    vcmd(idx) = min( CoefP'*H, wc_SAT) ;

elseif( wc <= -wc_DZ ) % CCW

    wc = max( wc, -wc_SAT ) ;
    H = [ wc^4 ; wc^3 ; wc^2 ; wc ; 1 ] ;

    vcmd(idx) = max( CoefN'*H, -wc_SAT) ;

else % Deadzone

    vcmd(idx) = 2.5 ; % stop voltage is issued to the motor

end

%-----
% EXPECTED LINEARIZATION RESULT
%-----
% Wcmd --> Linearization Function --> Vm --> Motor --> Wout
%-----

wout(idx) = interp1( Vm, wg, vcmd(idx) ) ;

end

%-----
% Plot Results
%-----

figure(1), plot(wg, Vm, 'b', bufwc, vcmd, 'r', 'Linewidth',2),
title('Linearization Function Design:  $V_m = f(\omega_c)$ '), 'FontSize',13)
legend({'static characteristics:  $V_m = g(\omega_{out})$ ',...
    'linearization function:  $V_m = f^{-1}(\omega_c)$ '}, 'FontSize',12)
grid on, box on, xlabel('wc [deg/s]', 'FontSize',12), ylabel('Vcmd [V]',
'FontSize',12)
set(gca, 'FontSize',12)

figure(2), plot(bufwc, wout, 'Linewidth',2),
title('Expected Linearization Result', 'FontSize',13)
legend({' $\omega_c \rightarrow f^{-1}(\omega_c) = V_m \rightarrow \text{motor } f(\cdot) \rightarrow \omega_{out}$ '}, 'FontSize',12)
grid on, box on, xlabel('wc [deg/s]', 'FontSize',12), ylabel('wout [deg/s]',
'FontSize',12)
set(gca, 'FontSize',12)

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